

Database Introduction

Learning Objectives

- Understand what a database is and its purpose
- Identify fields, records, and tables
- Recognize real-world uses of databases

Engage (5-7 minutes)

Think of a village voter list—it has names, addresses, ages, and ID numbers organized in rows and columns. A database is like a giant digital voter list that can store, search, and sort thousands of records instantly. Today we learn about this information organizing tool.

Explore (10-12 minutes)

Look at a simple database on the computer. See how data is organized in tables with rows and columns. Each column is a field (like “Name” or “Age”) and each row is a record (one person’s information). Try searching for a specific record and sorting the data by different fields.

Explain (10-12 minutes)

A database stores organized data that can be easily accessed and managed. Data is arranged in tables where columns are fields (types of information) and rows are records (individual entries). Databases let you search, sort, filter, and report data quickly. Examples include school records, library catalogs, and hospital patient lists.

Elaborate (8-10 minutes)

Create a simple database of class students with fields for Name, Roll Number, and Marks. Add at least five records. Practice searching for a student, sorting by roll number, and filtering by marks. Think of how your school office uses a database to track student information.

Evaluate (5-8 minutes)

Exit ticket: What is a database? Define field and record with examples. Give two real-world examples of databases used in Karnataka. Why is a database better than paper records for large amounts of information?